SUPPORTING ADULT GUIDANCE

Where two lines meet

Equations, Inequalities and Simultaneous Relationships · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson interprets the **intersection** of two lines as the **common solution** of a pair of simultaneous equations — the single (x, y) satisfying both. It is the graphical foundation for the algebraic methods (substitution and elimination) in the next two lessons, and it grounds 'solve simultaneously' in a picture.

Driving Question: *When must two relationships both be true — and where is that?*

The mathematics, in plain terms

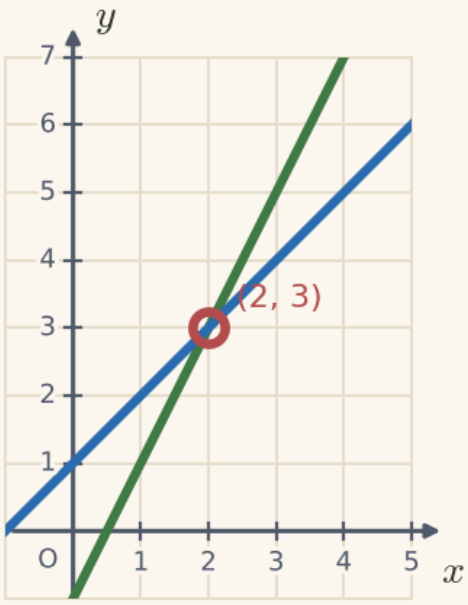
Each line is the solution set of its equation. A pair satisfying both equations therefore lies on both lines — at their intersection. Reading it off the graph gives an (approximate or exact) common solution; substituting it into each equation verifies it. Algebraically, equating the two expressions for y (valid precisely where the lines meet) yields x, and back-substitution gives y.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Reading the common solution

Where do $y = 2x - 1$ and $y = x + 1$ cross?

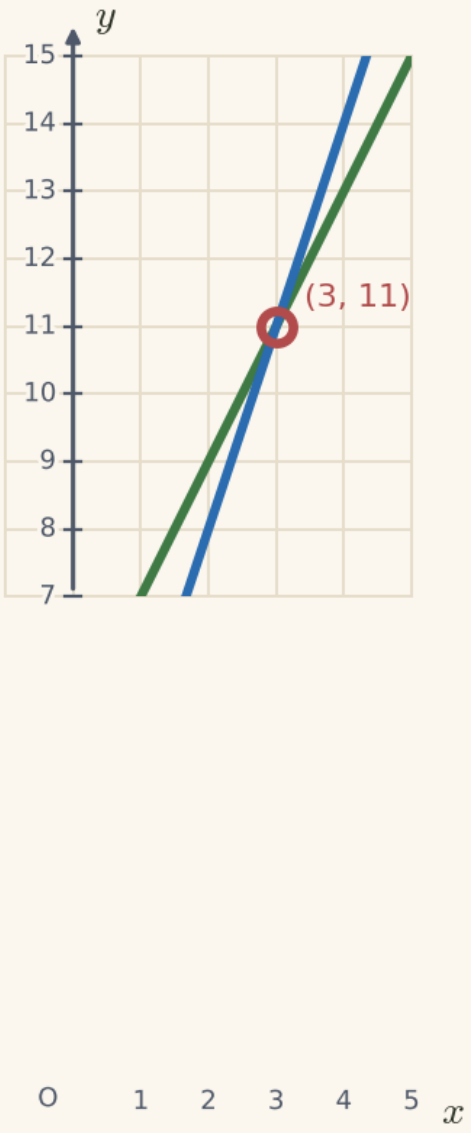


1. At the crossing the y-values agree, so set the sides equal.	$2x - 1 = x + 1$
2. Collect the x-terms and solve.	$x = 2$
3. Find y from either equation.	$y = 2 + 1 = 3$
4. Check in both.	$2\times2 - 1 = 3$ ✓ and $2 + 1 = 3$ ✓

(2, 3) — the crossing satisfies both equations, so it is the common solution.

Example 2 — A break-even point

Plan A costs $y = 2x + 5$; Plan B costs $y = 3x + 2$. When do they cost the same?



1. Set the two costs equal.	$2x + 5 = 3x + 2$
2. Solve for x (the hours).	$x = 3$
3. Find the shared cost.	$y = 2\times3 + 5 = 11$
4. Interpret the decision.	Same at (3, £11); cheaper plan differs either side

They break even at (3, £11) — the common solution. Before 3 hours one plan is cheaper; after, the other.

From two solution sets to one shared point

Build on Lesson 5: each line is a set of solutions, so the pair that works for both is where the sets overlap — the crossing. Read it, then *check in both* equations; a point that satisfies only one is a useful non-example. Introduce 'set the sides equal' as the algebraic shortcut once the graphical meaning is secure.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Read a crossing and prove it satisfies both equations.	~12 min
Movement	Step an x-line across to hunt where the two y-values agree.	~10 min
Reflection	Test points: only the crossing lies on both lines.	~10 min
Creativity	Find a break-even point where two plans cost the same.	~8 min
Rest	A short pause on two whole lines sharing one point.	~2 min
Nutrition	Intellectual: two conditions at once — 'which deal, and when?'.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading only one coordinate of the crossing.
A solution is a pair. Ask for both: 'x across and y up — what is the full point where they meet?'

Checking the crossing in one equation only.
Common means both. 'Does it satisfy the other equation too? Substitute and see.'

Thinking any point on a line solves the pair.
Only the shared point does. 'Is this point on the OTHER line as well? If not, it is not the common solution.'

Assuming lines always cross at whole numbers.
They need not — the stretch gives x = 0.5. The graph gives an estimate; equating the sides gives the exact value.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: reading a line as a set of solutions (Lesson 5), reading coordinates, and substituting a point. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner reads only the x of the crossing, or checks it in just one equation. The aim is a question that returns the reasoning.

- "What is the full point where they meet — both x and y?"
- "You checked it in one equation. Does it satisfy the other one too?"

- “Is this point on both lines, or only one? Which makes it common?”
- “Set the two right-hand sides equal — why is that true exactly at the crossing?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is reading off the graph accurate enough?

It gives a good estimate and the essential meaning. For an exact answer, set the two expressions for y equal and solve — which is exactly the substitution method of the next lesson.

What if the lines are parallel?

Then they never cross and there is no common solution — the equations are inconsistent. Worth mentioning if it arises; the focus here is the usual single-crossing case.

Why 'simultaneous'?

Because both equations must hold at the same time. The crossing is the one (x, y) that does that — the common solution.

How does this lead into the next lessons?

Setting the two right-hand sides equal is substitution; combining the equations to remove a variable is elimination. Both find this same crossing without drawing.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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